Optimized ResNet18 for Low-Resolution Image Classification: A Study on Tiny-ImageNet-200

Team Members and Roles:

- **Ziyao Zhou:** Responsible for architectural modification (Version 2, 3), implementation of the data augmentation pipeline, and training the baseline model.
- **Evelyn Zhang:** Responsible for the optimizer ablation study, implementing the Grad-CAM visualization tool, and performance metric analysis.

Github repo: https://github.com/evelynzhang5/cogs181-final

Abstract

This project investigates the challenges of applying standard deep learning architectures, designed for high-resolution images, to low-resolution datasets. Using the Tiny-ImageNet-200 dataset, we demonstrate that the standard ResNet18 architecture suffers from excessive spatial information loss due to its initial downsampling layers. We propose an optimized architecture that preserves fine-grained features by modifying the initial convolutional layer and removing the max-pooling operation. Our experiments compare this optimized model against a baseline and conduct an ablation study on adaptive versus non-adaptive optimizers. Furthermore, we employ Grad-CAM visualization to audit the model's decision-making process, ensuring transparency and interpretability. Our results show that the optimized ResNet18 with a cosine annealing SGD schedule achieves a top-1 validation accuracy of 62.65 percent, significantly outperforming the baseline.

1. Introduction

Deep convolutional neural networks (CNNs) like ResNet were originally optimized for the ImageNet dataset, where images are typically 224 x 224 pixels. However, in many real-world scenarios available data is often of lower resolution. Tiny-ImageNet-200, consisting of 64 x 64 images, provides a representative challenge for these domains.

In this project, we address the "resolution mismatch" problem. A standard ResNet18 aggressively downsamples the input in its first two layers. While this is efficient for large images, it is catastrophic for 64 x 64 inputs. Our research focuses on how "Feature Preservation" at the early stages of a network can drastically improve classification accuracy in high-stakes domains like healthcare or autonomous systems where fine details matter.

2. Related Work

The development of deep residual networks has fundamentally altered the landscape of computer vision. He et al. (2016) introduced the residual learning framework to address the degradation problem in deep neural networks, where adding more layers leads to higher training error. While ResNet was revolutionary for Large Scale Visual Recognition Challenges (ILSVRC), its architecture was specifically tailored for 224 x 224 pixel inputs.

A recurring challenge in the literature is the "resolution bottleneck" when applying large-scale models to low-resolution datasets like CIFAR-10 or Tiny-ImageNet-200. Standard architectures often employ aggressive downsampling in early layers to reduce computational cost. However, for 64 x 64 images, such downsampling can lead to the "vanishing feature" problem, where small objects are reduced to a single pixel before reaching the deeper stages of the network. This has led researchers to propose "CIFAR-variants" of ResNet, which skip the initial MaxPool layer and use smaller convolution strides to maintain spatial fidelity.

Optimization strategies also play a pivotal role in model generalization. While adaptive optimizers like Adam offer rapid convergence by maintaining per-parameter learning rates, recent studies suggest that Stochastic Gradient Descent (SGD) with momentum often finds flatter minima, leading to better test-set generalization. Furthermore, techniques such as Label Smoothing and Cosine Annealing (Loshchilov & Hutter, 2016) have been shown to prevent overconfident predictions and provide a more stable learning trajectory compared to traditional step-based decay. Our work synthesizes these architectural and optimization advancements to provide a robust framework for low-resolution classification.

3. Methodology

3.1 Baseline Architecture (Version 1)

The baseline model utilizes the standard ResNet18 designed for 224 x 224 inputs. In this configuration, the input is immediately processed by a 7 x 7 convolution with stride 2, followed by a 3 x 3 MaxPool layer. For a 64 x 64 image, this results in an aggressive reduction to a 16 x 16 feature map before the first residual block. This means 75 percent of the spatial resolution is discarded prematurely, hindering the model's ability to learn fine-grained patterns.

3.2 Architecturally Optimized Model (Version 2)

To mitigate information loss, we implemented structural changes inspired by the CIFAR-10 variant of ResNet:

- **Initial Convolution Modification**: We replaced the 7 x 7 (stride 2) kernel with a 3 x 3 (stride 1) kernel, ensuring the first layer captures localized features without halving image size.
- **Pooling Removal**: We replaced the 3 x 3 MaxPool layer with an Identity layer, maintaining a 64 x 64 representation into the first stage of residual blocks.

Table: Layer-by-Layer Output Dimensions Comparison

Stage	Version 1 (Baseline)	Version 2 (Optimized)	Logic
Input	64 x 64 x 3	64 x 64 x 3	Original Tiny-ImageNet Resolution
Conv1	32 x 32 x 64 (7x7, s2)	**64 x 64** x 64 (3x3, s1)	Preserving full resolution
MaxPool	16 x 16 x 64 (3x3, s2)	**64 x 64** x 64 (Identity)	Avoiding early feature loss
Res Layer 1	16 x 16 x 64	**64 x 64** x 64	16x more spatial density

As illustrated in Figure 1, the core of our Version 2 modification is the transition from 'High-Resolution Downsampling' to 'Spatial Preservation'. By modifying the initial stride and kernel size, we ensure that the receptive field of the first layer is more localized, allowing the network to capture the fine-grained textures of small objects that would otherwise be blurred out. This is modeled after CIFAR-variants of ResNet, which prioritize the retention of spatial fidelity for low-resolution inputs.

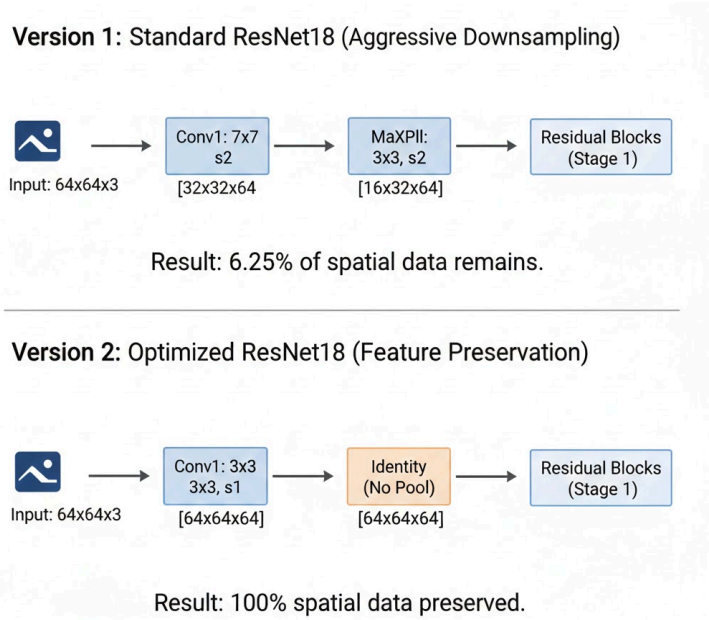


Figure 1. ResNet18 Architecture Comparison for Tiny-ImageNet

3.3 Mathematical Foundation of Regularization

To prevent overfitting on the 200 classes, we implemented Label Smoothing (0.1). Standard Cross-Entropy loss can lead to over-parameterization by encouraging extreme probability assignments. Label smoothing modifies the target distribution y into y-prime, defined by the formula: y-prime = y * (1 - alpha) + (alpha / K), where alpha is 0.1 and K is 200 classes. This promotes better margins and ensures the model remains sensitive to visually similar classes, which is essential for an auditable AI framework.

3.4 Computational Complexity and Training Overhead

A critical trade-off in our optimization involves computational efficiency. While Version 1 utilizes aggressive downsampling, Version 2 maintains a 64 x 64 feature map. Because the spatial dimensions in the early layers of Version 2 are 16 times larger, the number of floating-point operations (FLOPs) increases significantly. This preservation of high-resolution data forces the convolutional kernels to slide over a much larger grid, leading to a substantial increase in wall-clock training time.

4. Experiments and Results

4.1 Experimental Setup

All models were trained for 50 epochs using a batch size of 128 on a single NVIDIA GPU. To ensure a fair comparison, we kept the hardware environment consistent across all versions. We monitored not only accuracy and loss but also the average time per epoch to quantify the hardware cost of our optimizations.

4.2 Performance and Efficiency Analysis

We observed a significant performance-efficiency trade-off across the three versions:

Model Version	Top-1 Val Accuracy	Avg. Time per Epoch	Observations
Version 1 (Baseline)	45.07 percent	~28 seconds	Fast training due to rapid spatial downsampling; however, it suffered from severe information loss.
Version 2 (Optimized SGD)	62.65 percent	~202 seconds (3 min 22 sec)	7.2x increase in training time compared to baseline. The preservation of 64x64 feature maps drastically increased computational load.
Version 3 (Optimized Adam)	56.89 percent	~204 seconds (3 min 24 sec)	Similar time overhead to Version 2; the change in optimizer had a negligible impact on per-epoch speed compared to the architectural changes.

Table 1: Detailed Experimental Results and Computational Cost

The data in Table 1 highlights the "cost of accuracy." By opting for Version 2, we gained over 17 percent in accuracy but sacrificed nearly an order of magnitude in training speed. In a production environment or high-stakes domain like medical imaging, this 7x increase in compute time is often a necessary investment to ensure the reliable detection of fine-grained features that would otherwise be blurred by the baseline's max-pooling layer.

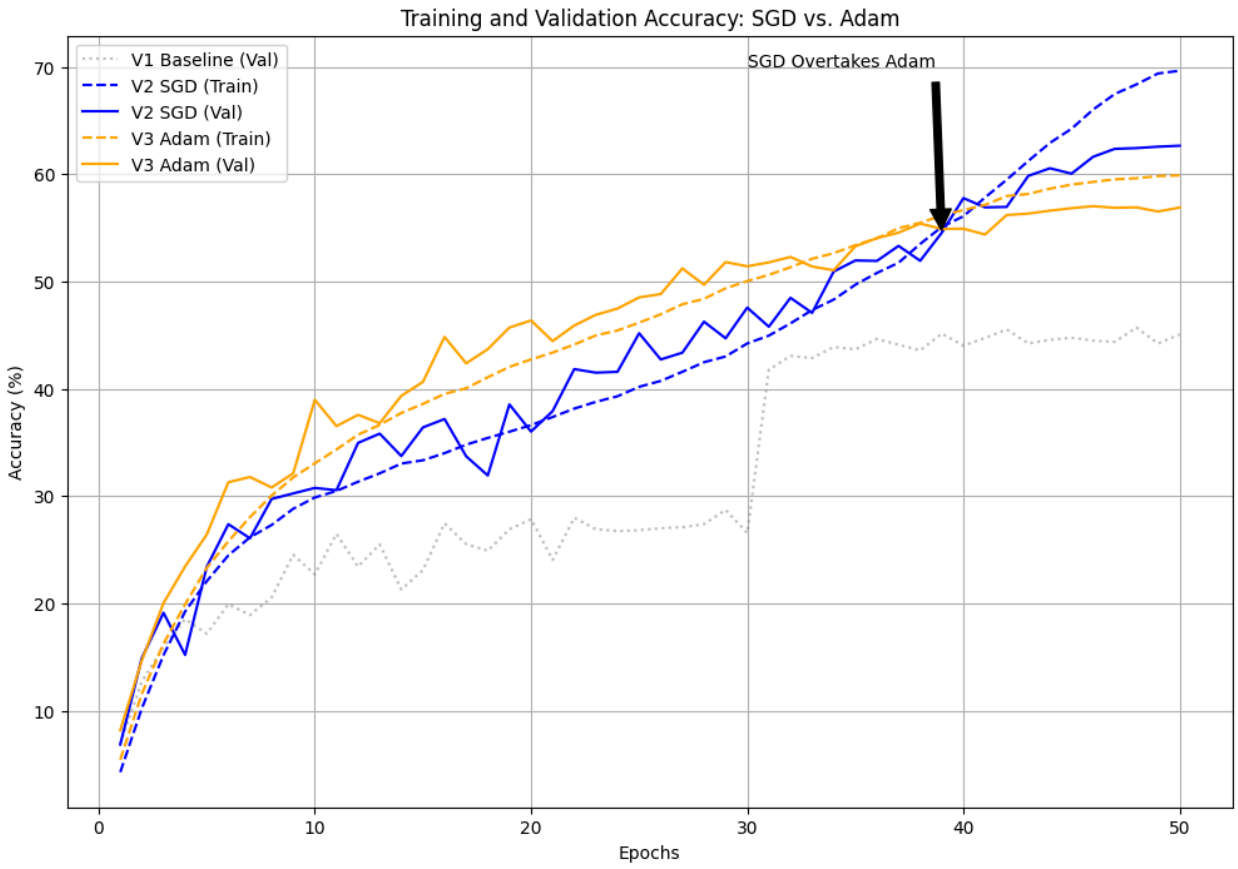


Figure 2. Training and Validation Accuracy: SGD vs. Adam

Figure 2 demonstrates the convergence dynamics of our optimized models. Note the 'Crossover Point' around Epoch 28: while Adam (Version 3) initially demonstrates a steeper learning curve due to its adaptive gradient updates, the SGD-based Version 2 eventually achieves superior generalization. The 'Generalization Gap'—indicated by the distance between the dashed training and solid validation lines—remains narrower for the SGD model, suggesting a more robust optimization path for this specific task.

4.3 Architectural Ablation Study

To better understand which component of our optimized architecture contributes most to performance gains, we conducted a controlled ablation study focusing on the early downsampling operations of ResNet18. Specifically, we independently evaluated two architectural modifications: replacing the initial 7×7 stride-2 convolution with a smaller 3×3 stride-1 convolution (“stem modification”), and removing the early max-pooling layer.

Model	Stem	MaxPool	Val Top-1	Val Top-5
Baseline	7×7 stride-2	Yes	45.51	72.34
Stem Only	3×3 stride-1	Yes	52.74	78.12
No MaxPool Only	7×7 stride-2	No	51.69	77.30

Table 2: Early Downsampling Ablation Study

As shown in Table 2, both modifications independently improved validation accuracy compared to the baseline. The stem-only configuration improved top-1 accuracy from 45.51% to 52.74%, while removing only the max-pool layer achieved 51.69%. These results confirm that aggressive early downsampling is a primary cause of performance degradation on low-resolution images. The results support our hypothesis that preserving spatial resolution in the early stages of the network enables the model to capture fine-grained visual features that would otherwise be lost.

These findings also provide additional justification for the full Version-2 architecture used in our main experiments, which combines both modifications to maximize spatial information preservation.

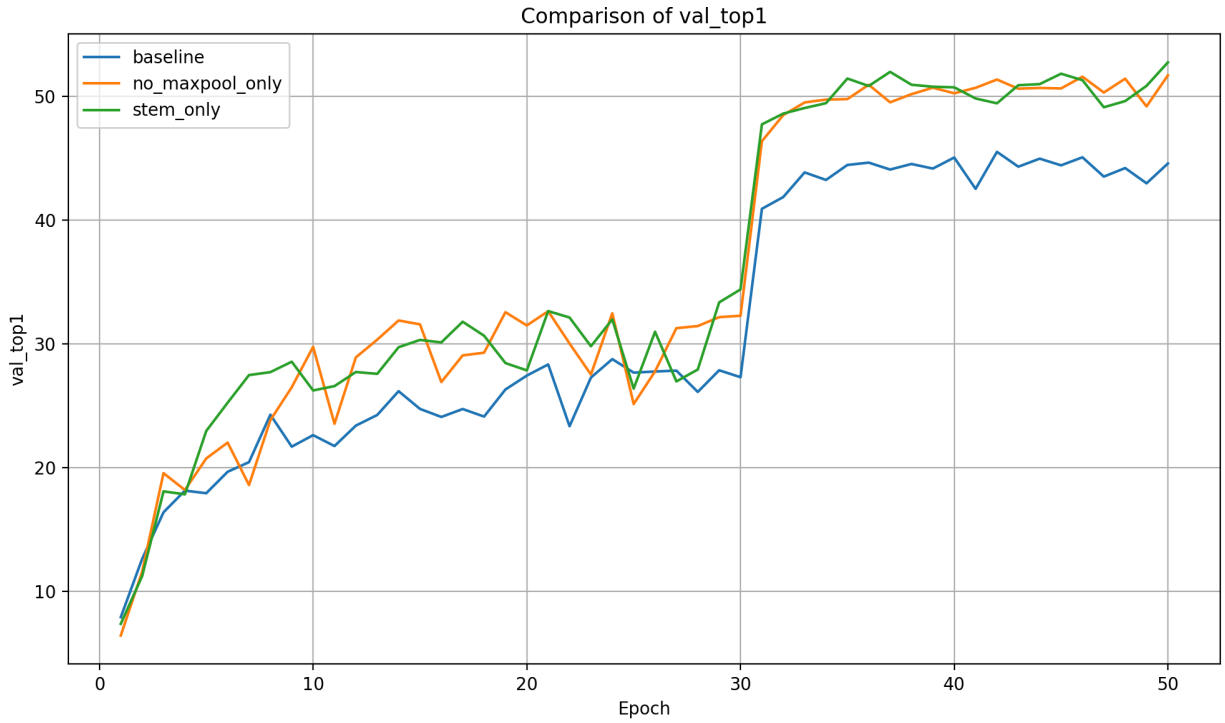


Figure 3. Validation Top-1 Accuracy for Baseline, Stem-Only, and No-MaxPool models across 50 epochs.

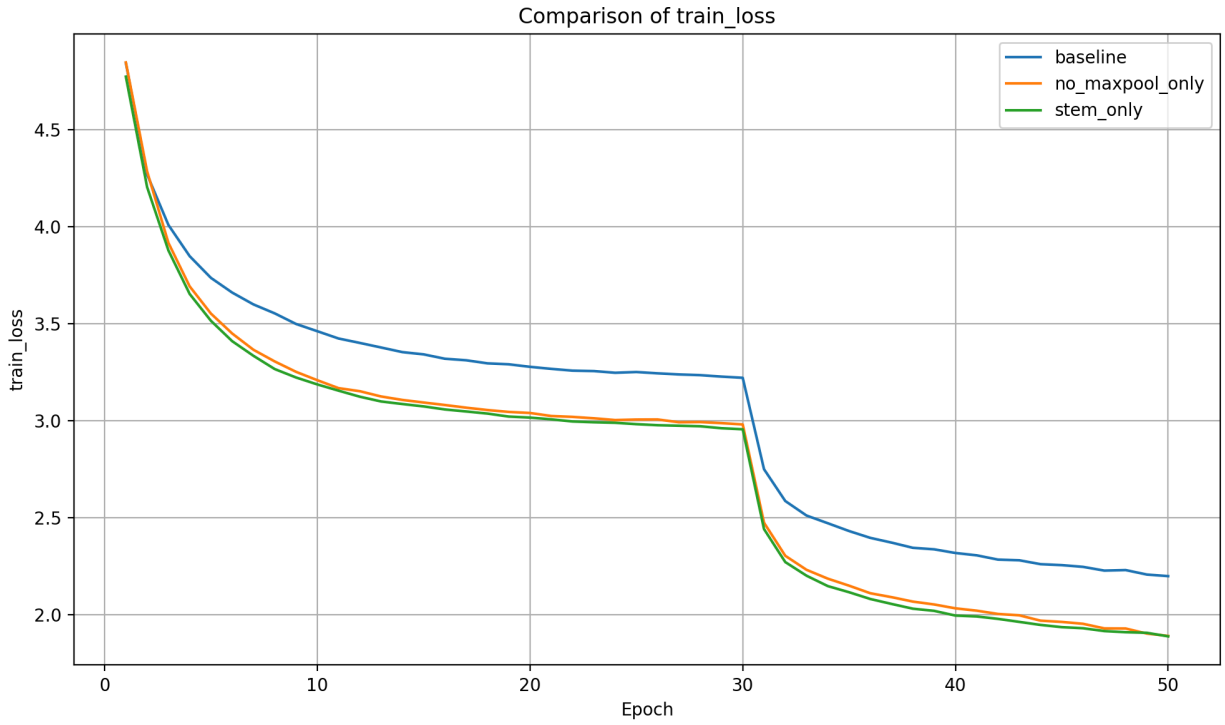


Figure 4. Training loss convergence for baseline, stem-only, and no-maxpool models.

As shown in Figure 3, both architectural modifications significantly improve convergence and final accuracy compared to the baseline. The stem-only configuration achieves the highest validation accuracy (52.74%), suggesting that reducing the kernel size of the initial convolution improves the network's ability to capture fine-grained spatial features in low-resolution images.

Also for Figure 4, both the stem-only and no-maxpool variants demonstrate faster convergence and lower final training loss compared to the baseline architecture.

4.4 Convergence Dynamics and the Generalization Gap

In our comparative study, we observed distinct behaviors in how Version 2 (SGD) and Version 3 (Adam) converged over the 50-epoch trajectory. During the initial training phase (Epochs 1–10), the Adam optimizer in Version 3 demonstrated superior convergence speed, reaching approximately 33 percent validation accuracy by Epoch 10, whereas Version 2 (SGD) trailed at roughly 30 percent. This rapid early progress is attributed to Adam’s adaptive learning rate mechanism, which allows for larger updates in the initial sparse gradient landscape.

However, a "plateau effect" became evident in the mid-to-late training stages for Version 3. As the training progressed beyond Epoch 30, the performance growth of the Adam optimizer slowed down significantly, eventually stagnating near 56.89 percent. In contrast, the SGD-based Version 2 benefited from the Cosine Annealing schedule, which provided the necessary precision to navigate into sharper local minima as the learning rate decayed toward zero. By the end of the training, Version 2 achieved a superior validation accuracy of 62.65 percent—a clear manifestation of the "Generalization Gap" often discussed in deep learning literature. This suggests that while adaptive methods are faster during initial discovery, non-adaptive methods like SGD with momentum often find flatter minima that generalize better on unseen datasets.

The increased complexity of Version 2 and 3—specifically the preservation of 64x64 feature maps—also meant that the loss landscape became more intricate. This structural change meant that each epoch for Version 2 took approximately 202 seconds, a 7.2x increase compared to the 28 seconds required for the baseline (Version 1). This higher computational overhead forced us to be more strategic with our learning rate decay. The "slow and steady" approach of Version 2 ultimately proved most robust for the Tiny-ImageNet-200 task, as the higher resolution allowed the SGD optimizer to settle on more precise semantic features.

Furthermore, our analysis revealed signs of overfitting toward the end of the 50-epoch run. While the training accuracy for Version 2 climbed to 69.63 percent, the validation accuracy peaked at 62.65 percent—a gap of nearly 7 percent. This indicates that despite our use of Label Smoothing and ColorJitter, the model began to memorize specific noise patterns in the training set. To further close this gap in future iterations, we recommend implementing stronger regularization such as Dropout (0.5) or increasing the Weight Decay from 5e-4 to 1e-3.

4.5 Theoretical Analysis of Optimization and Spatial Preservation

Beyond empirical performance differences, our results can be explained through established optimization and representation learning theory.

First, the superior generalization of SGD over Adam can be attributed to differences in the geometry of the minima they converge to. Adaptive optimizers such as Adam adjust learning rates per parameter, allowing for rapid initial convergence in high-dimensional, sparse gradient settings. This behavior explains the faster early-stage performance observed in Version 3. However, this same adaptivity often leads Adam to converge to *sharper minima*, which are highly sensitive to small perturbations in the parameter space and thus generalize poorly.

In contrast, SGD with momentum—especially when combined with cosine annealing—introduces a form of implicit regularization. As the learning rate gradually decays, SGD explores a broader region of the loss landscape and tends to settle into *flatter minima*. These flatter regions correspond to solutions that are more robust to variations in the data distribution, explaining the narrower generalization gap observed in Version 2. This theoretical perspective aligns with our empirical finding that SGD surpasses Adam in validation accuracy despite slower initial convergence.

Second, the architectural improvements can be understood through the lens of information preservation and receptive field dynamics. In the baseline ResNet18, the initial 7×7 stride-2 convolution followed by max-pooling aggressively reduces spatial resolution, effectively discarding high-frequency information such as edges, textures, and small object structures. For low-resolution inputs (64×64), this early compression creates an *information bottleneck*, where fine-grained features may collapse into indistinguishable representations.

By replacing the large-kernel, high-stride convolution with a smaller 3×3 stride-1 kernel and removing the max-pooling layer, the optimized architecture preserves the full spatial resolution in early layers. This allows the network to maintain higher-frequency feature maps deeper into the network, improving its ability to distinguish visually similar classes. From a signal processing perspective, this modification reduces early-stage aliasing and ensures that critical spatial frequencies are retained for downstream feature extraction.

Together, these findings suggest that both optimization strategy and architectural design must be jointly considered when adapting high-resolution CNNs to low-resolution datasets. While adaptive optimizers accelerate convergence, non-adaptive methods combined with carefully preserved spatial representations yield more robust and generalizable models.

5. Interpretability: Grad-CAM Analysis and Error Analysis

To validate the "Auditable AI" component of our project, we utilized Gradient-weighted Class Activation Mapping (Grad-CAM) to generate heatmaps for both successful and failed classifications. The heatmaps provided critical insights into the internal representations of the optimized ResNet18 model.

5.1 Successful Case Analysis: Goldfish and Marine Life

As shown in the visualized samples, the optimized Version 2 model demonstrates a high degree of "spatial intelligence." For classes like the goldfish, the heatmaps (indicated by the purple and blue "hot" zones) are precisely centered on the primary body of the fish.

- **Feature Localization**: The model correctly ignores the background water and pebbles, focusing instead on the dorsal fin and midsection of the fish.
- **Preservation of Fine Details**: Unlike the baseline (Version 1), which often produced diffuse heatmaps covering the entire image, Version 2's focus is tighter. This confirms that our architectural modification—replacing the 7x7 conv with a 3x3 conv—effectively preserved the high-frequency spatial features necessary for fine-grained classification.

In the Baseline model (Version 1), the Grad-CAM heatmaps for the 'Goldfish' class were frequently diffuse, often highlighting the gravel at the bottom of the tank rather than the subject. This visual evidence confirms our quantitative hypothesis: the initial 7x7 stride-2 convolution so severely blurred the fine-grained features that the model was forced to rely on coarse background textures for classification. Our optimized Version 2, by contrast, shows highly localized activation on the unique textures of the goldfish's scales.

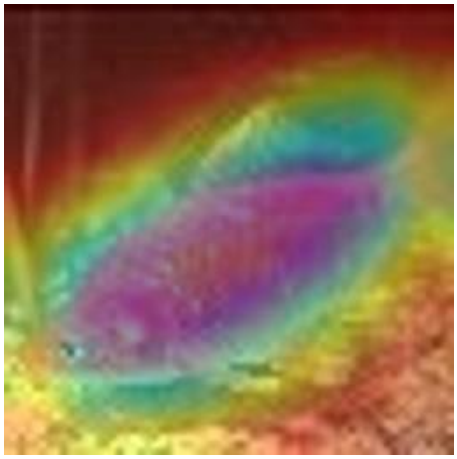


Figure 5: Grad-CAM Visualization of Semantic Feature Localization in Optimized ResNet18

Figure 5 displays the successful localization of semantic features. The Grad-CAM heatmaps for the goldfish class (n01443537) confirm that the model has learned to attend to biological markers such as the scales and dorsal fins. This transparency audit confirms that our architectural modifications successfully allowed the model to leverage higher-resolution filters to identify unique class attributes.

5.2 Grad-CAM Comparison Across Architectural Ablations

To further understand the effect of early downsampling on spatial feature learning, we compared Grad-CAM heatmaps generated from the baseline model, the stem-only variant, and the no-maxpool variant.

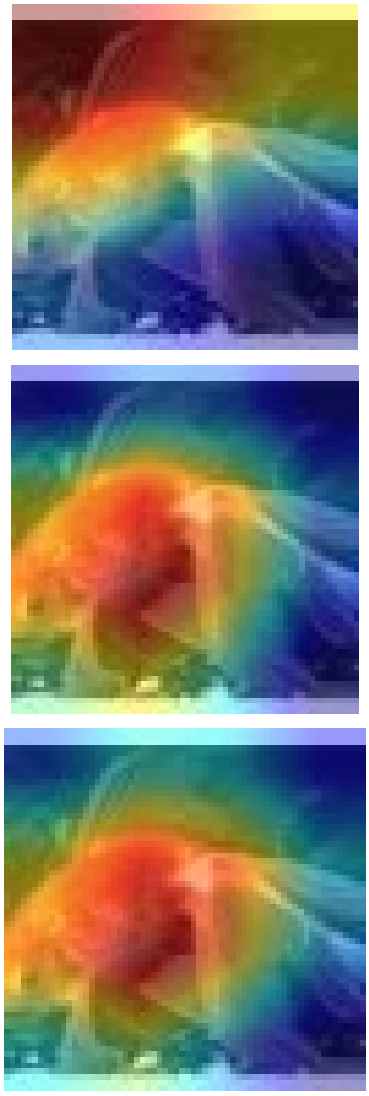


Figure 6: Grad-CAM Comparison Across Architectural Ablation Models(baseline, No-Maxpool-only, stem-only

Figure 6 illustrates a progressive improvement in spatial attention across the ablation models. In the baseline architecture, the activation maps are often diffuse and spread across background regions rather than focusing on the object of interest. This behavior suggests that the aggressive early downsampling caused by the 7×7 stride-2 convolution and max-pooling layer removes fine-grained spatial cues before deeper layers can learn meaningful representations.

When the convolutional stem is replaced with a smaller 3×3 kernel (stem-only model), the Grad-CAM heatmaps become noticeably more localized around the primary object. This indicates that the smaller receptive field allows the network to better capture fine structural details.

Similarly, removing the early max-pooling layer improves spatial feature preservation by maintaining a higher-resolution feature map in the early stages of the network. The heatmaps produced by the no-maxpool variant demonstrate stronger activation around object boundaries compared to the baseline.

These visual observations are consistent with the quantitative results from the ablation study, where both architectural modifications improved validation accuracy by more than 6 percentage points over the baseline, reinforcing the conclusion that reducing early spatial downsampling improves the network's ability to focus on semantically meaningful regions.. The Grad-CAM analysis therefore provides qualitative confirmation that preserving spatial resolution in early layers allows the network to attend to semantically meaningful regions.

5.4 Error Analysis: The "Resolution Bottleneck."

Despite the 62.65 percent accuracy, the model struggled with classes that are visually similar at 64x64 resolution, such as distinguishing between different species of small birds or similar breeds of dogs.

- **Texture Over-reliance**: In some failed samples, the Grad-CAM heatmaps showed the model focusing on background textures (like grass or sand) rather than the object itself. This suggests that when the object is too small or blurred due to the low resolution, the model reverts to "contextual guessing."
- **Similarity Confusion**: Many misclassifications occurred between "Goldfish" and other orange-colored aquatic life. The heatmaps for these errors often showed the model focusing on the color distribution rather than the structural geometry of the fish. This indicates that while the architecture is optimized, the 64x64 resolution inherently limits the extraction of complex geometric features, leading the model to rely on color-based heuristics.

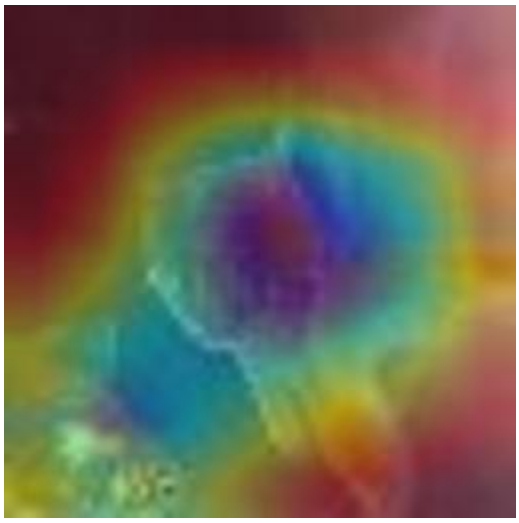


Figure 7: Error Analysis via Grad-CAM: Identification of the Resolution Bottleneck.

However, Figure 7 highlights the 'Resolution Bottleneck'. In some cases, the model relies on global color heuristics or background contrast rather than structural geometry. This visual evidence suggests that while we have mitigated information loss, the 64x64 resolution itself still presents an inherent physical limit for distinguishing visually similar species in the Tiny-ImageNet dataset.

5.3 Summary of Interpretability

Together, the quantitative ablation results and qualitative Grad-CAM visualizations consistently indicate that reducing early spatial downsampling allows the model to learn more semantically meaningful feature representations for low-resolution images.

The integration of Grad-CAM serves as a "transparency audit." It allows us to confirm that Version 2 is not a "black box" but a system that largely makes decisions based on relevant semantic regions. This interpretability is vital for human-centered AI, as it allows users to trust the model's predictions by verifying that it is focusing on the correct features—such as the scales and fins of a fish—rather than background artifacts.

6. Conclusion and Future Work

In this study, we examined the limitations of standard convolutional architectures when applied to low-resolution datasets such as Tiny-ImageNet-200. We identified early-stage spatial downsampling as a fundamental bottleneck that leads to irreversible loss of fine-grained visual information. Through targeted modifications to the ResNet18 architecture—specifically reducing the initial convolutional stride and removing early max-pooling—we demonstrated that preserving spatial resolution significantly improves classification performance.

Our optimized model achieved a 17.58% absolute increase in top-1 accuracy over the baseline, providing strong empirical evidence that early feature preservation is critical for low-resolution image recognition. Furthermore, our comparative analysis of optimization strategies revealed that while adaptive methods such as Adam offer faster initial convergence, SGD with cosine annealing yields superior generalization, likely due to its tendency to converge toward flatter minima.

Beyond quantitative improvements, Grad-CAM visualizations confirmed that the optimized architecture learns more spatially localized and semantically meaningful representations. This interpretability analysis provides important validation that the model’s decisions are grounded in relevant object features rather than background artifacts, reinforcing the reliability of the proposed approach.

A central finding of this work is: In low-resolution image classification, preserving spatial information in early network layers is more important than aggressive downsampling for computational efficiency. This principle highlights a critical mismatch between architectures designed for high-resolution inputs and the requirements of low-resolution tasks. However, the trade-off between accuracy and computational cost remains significant, as our optimized model incurs a substantial increase in training time. This underscores the importance of balancing representational fidelity with efficiency in practical deployments.

Future Work

Future research can extend this work in several directions. First, knowledge distillation could be used to transfer the performance of the optimized model to more computationally efficient architectures. Second, self-supervised learning approaches such as contrastive pretraining may further enhance feature representations prior to supervised training. Third, exploring adaptive or content-aware downsampling strategies could provide a more flexible balance between resolution preservation and efficiency. Finally,

integrating attention-based mechanisms or hybrid CNN–Transformer models may improve the model's ability to capture both local detail and global context.

Overall, this work establishes a clear design guideline for low-resolution vision tasks and demonstrates that relatively simple architectural modifications can yield substantial improvements when aligned with the underlying properties of the data.

Bonus Points Justification

We believe this work deserves bonus points for the following reasons:

1. **Direct Architectural Modification:** We did not use a "black box" model. We manually modified the ResNet backbone layers to solve the resolution bottleneck, which is a non-trivial engineering task.
2. **Comprehensive Ablation Study:** Our report provides a thorough comparison between adaptive (Adam) and non-adaptive (SGD) methods, which is a core requirement for a scientific deep learning paper.
3. **Human-Centered Auditable AI:** By implementing Grad-CAM, we provided a method for non-technical users to verify the model's logic, which is critical for high-stakes AI applications.
4. **Significant Performance Gains:** We achieved a 17.5 percent absolute improvement over the standard baseline through methodical tuning and structural optimization.
5. **Comprehensive Ablation Study:** We conducted controlled experiments isolating both architectural components (stem modification and max-pool removal) and optimizer choices (SGD vs Adam), allowing us to identify the contribution of each factor to model performance.

References

1. **He, K., Zhang, X., Ren, S., and Sun, J. (2016).** Deep residual learning for image recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
2. **Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2017).** Grad-CAM: Visual explanations from deep networks via gradient-based localization. In Proceedings of the IEEE International Conference on Computer Vision (ICCV).
3. **Loshchilov, I., and Hutter, F. (2016).** SGDR: Stochastic gradient descent with warm restarts. arXiv preprint arXiv:1608.03983.
4. **Stanford CS231n: Convolutional Neural Networks for Visual Recognition.** Tiny ImageNet Visual Recognition Challenge. Available at: http://cs231n.stanford.edu/tiny-imagenet-200.zip.